MILITARY SPECIFICATION

COAXIAL ASSEMBLIES, MLT TYPE (0.03 to 5.5 GHz)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for a series of MLT type coaxial assemblies operating over the frequency range of 0.03 to 5.5 GHz.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-87104	-Coaxial Assemblies, General Specification For
MIL-C-87104/6	-Coaxial Assemblies, LT Type (0.03 to 5.5 GHz)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification applicability. All requirements of MIL-C-87104 shall apply to this specification with modifications as specified herein. Modifications (options, specific values, exceptions, or additions) to MIL-C-87104 shall be identified by similarity of paragraph headings herein. When the general specification and this document conflict, the requirements of this document shall govern.

FSC 5985

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Qualification. This specification requires qualification inspection.

3.3 Parts, materials, and processes

3.3.1 Finishes

3.3.1.1 Plating

3.3.1.1.1 The center contacts on connectors and receptacles shall be silver-plated.

3.3.1.1.2 The outer contacts on male connectors and the inner contact surfaces of the outer contact on female connectors and receptacles shall be silver-plated.

3.3.1.2 Dimensions. All dimensions specified on the figures herein shall apply after plating.

3.3.1.3 Sharp edges. All sharp edges outlined on the figures herein shall be broken 0.015 inch.

3.3.1.4 Surface roughness. Unless otherwise indicated on the figures herein, surface roughness on all machined surfaces shall be 63.

3.3.2 Marking

3.3.2.1 Tags A and B information. The information contained on tags A and B, referenced on the figures herein, shall be as required by the contract (see 6.1.1 herein).

3.4 Design and construction

3.4.1 Functional subassemblies, form factor, and interface design. The functional subassemblies, form factor, and interface design requirements for the coaxial assemblies shall be in accordance with table I herein.

3.5 Performance

3.5.1 Electrical properties. Specific values for the electrical properties of the coaxial assemblies shall be as specified herein.

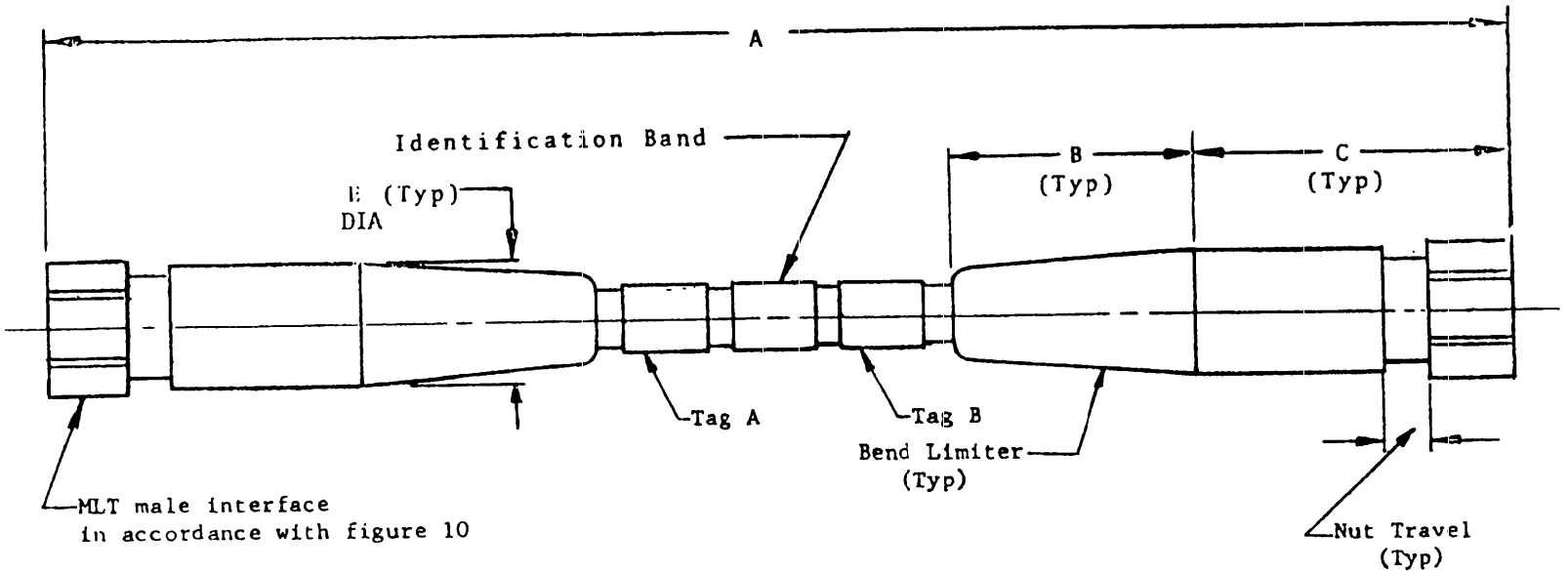
3.5.1.1 Frequency range. The coaxial assemblies shall operate over the frequency range of 0.03 to 5.5 GHz.

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TABLE I. Design and construction paramters.

COAXIAL ASSEMBLIES	FORM FACTOR	MALE INTERFACE	FEMALE INTERFACE
CABLE ASSEMBLY CG-3795/U (Type A)	Figure 1	Figure 10	
CABLE ASSEMBLY CG-3792/U (Type B)	Figure 2	Figure 10	
CABLE ASSEMBLY CG-3802/U (Type C)	Figure 3	Figure 10	Figure 12
CABLE ASSEMBLY CG-3803/U (Type D)	Figure 4	Figure 10	Figure 12
RIGHT ANGLE ADAPTER	Figure 5	Figure 10	Figure 12
PANEL RECEPTACLE	Figure 6		Figure 12
RIGHT ANGLE TRANSITION ADAPTER	Figure 7	(Figure 10 of (MIL-C-87104/6)	Figure 12
STRAIGHT TRANSITION ADAPTER	Figure 8	(Figure 10 of (MIL-C-87104/6)	Figure 12
BULKHEAD ADAPTER	Figure 9		Figure 12

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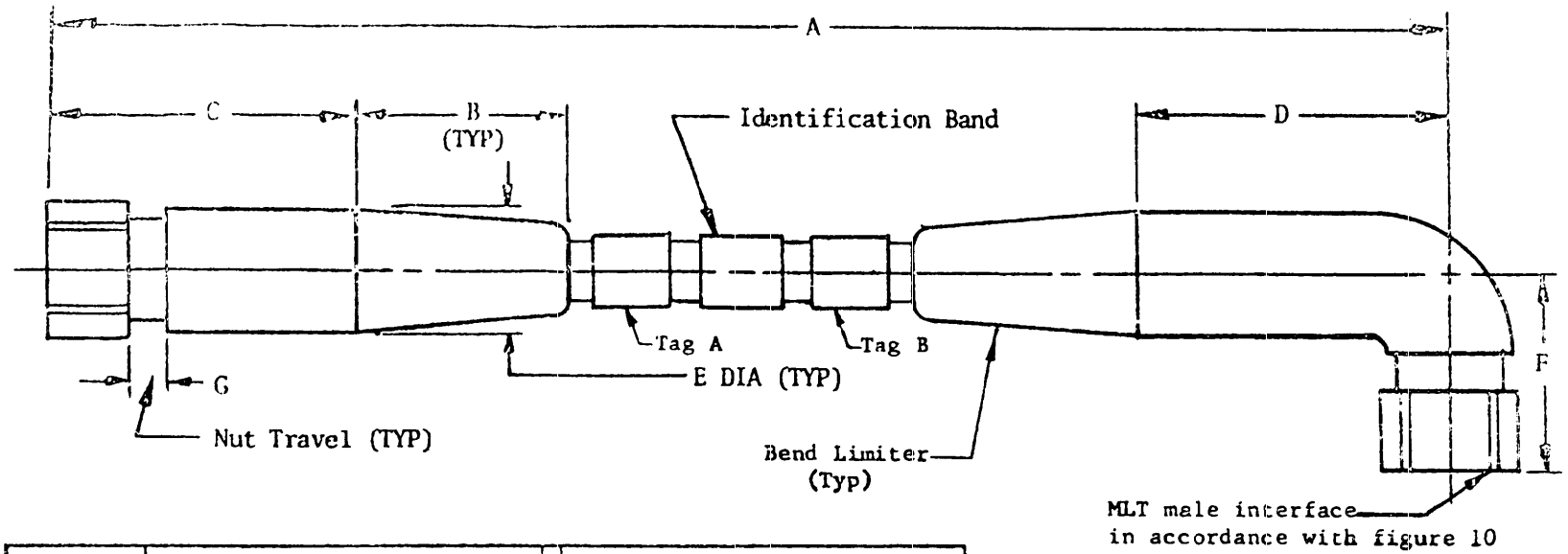
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	as required	note 1	as required	note 1
B	3.000	±.063	76.2	±1.6
C	3.750	±.063	95.25	±1.6
D	.500	±.031	12.7	±0.8
E	1.500	max	38.1	max

NOTES:

1. Tolerance for over-all cable length is ±0.25 inch up to 100 inches, and an additional ±0.25 inch for each additional 100 inches in excess of the first 100 inches.
2. Tags A and B to identify mating connector.

FIGURE 1. MLT cable assembly (Type A).

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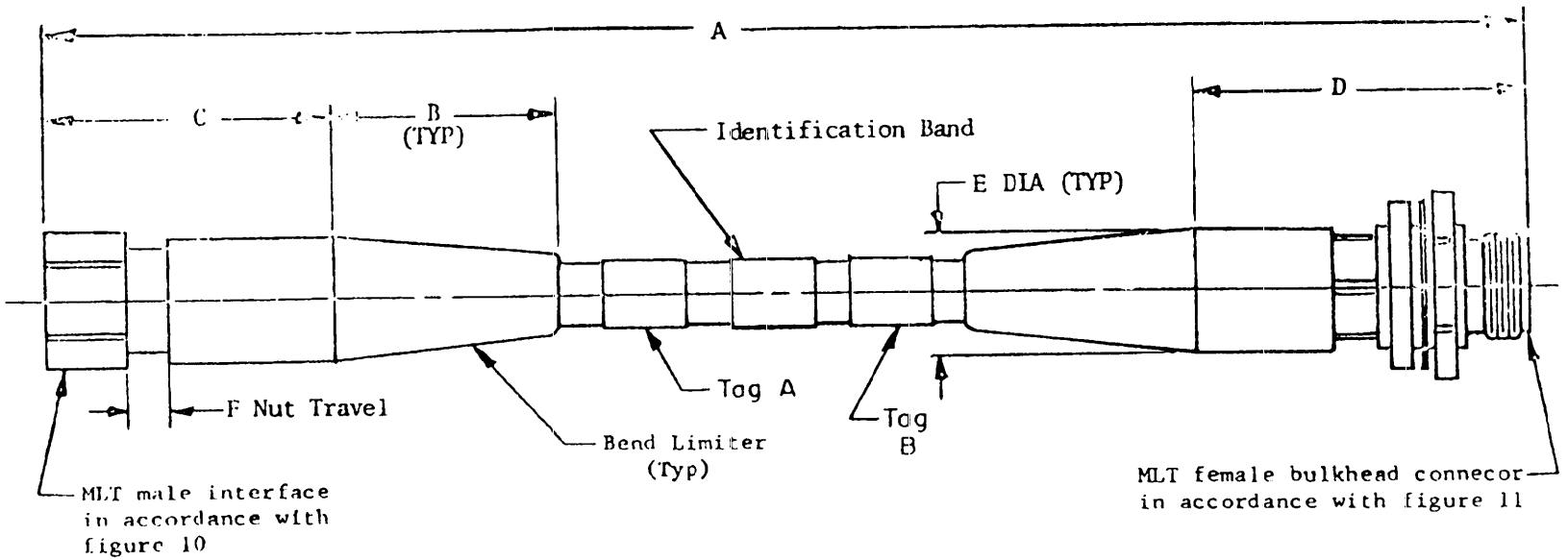
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	as required	note 1	as required	note 1
B	3.0	±0.06	76.2	±1.6
C	3.75	±0.06	95.25	±1.6
D	3.94	±0.06	100.01	±1.6
E	1.50	max	38.10	max
F	2.41	±0.06	61.12	±1.6
G	0.5	±0.03	12.70	±0.8

NOTES:

1. Tolerance for over-all cable length is ±0.25 inch up to 100 inches, and an additional ±0.25 inch for each additional 100 inches in excess of the first 100 inches.
2. Tags A and B to identify mating connector.

FIGURE 2. MLT cable assembly (Type B).

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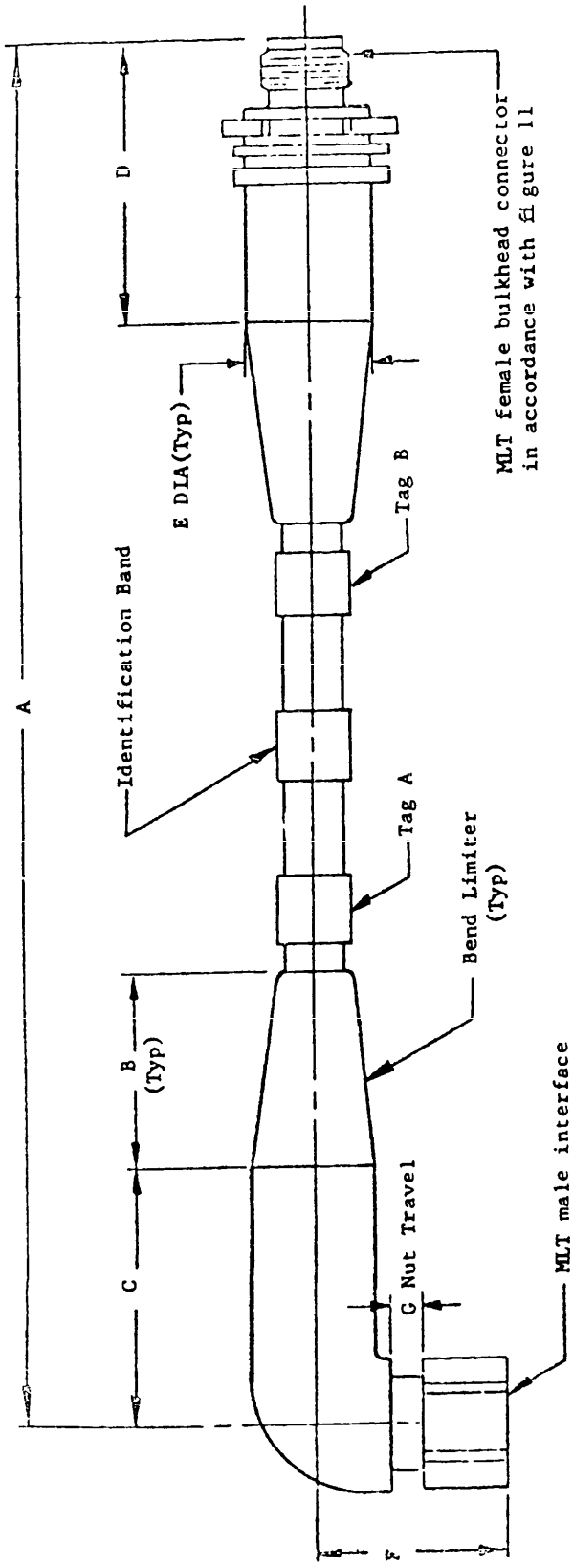
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	as required	note 1	as required	note 1
B	3	±.06	76.2	±1.6
C	3.75	±.06	95.25	±1.6
D	4	max	101.6	max
E	1.5	max	38.1	max
F	0.5	±0.03	12.7	±0.8

NOTES:

1. Tolerance for over-all cable length is ±0.25 inch up to 100 inches, and an additional ±0.25 inch for each additional 100 inches in excess of the first 100 inches.
2. Tags A and B to identify mating connector.

FIGURE 3. MLT cable assembly (Type C).

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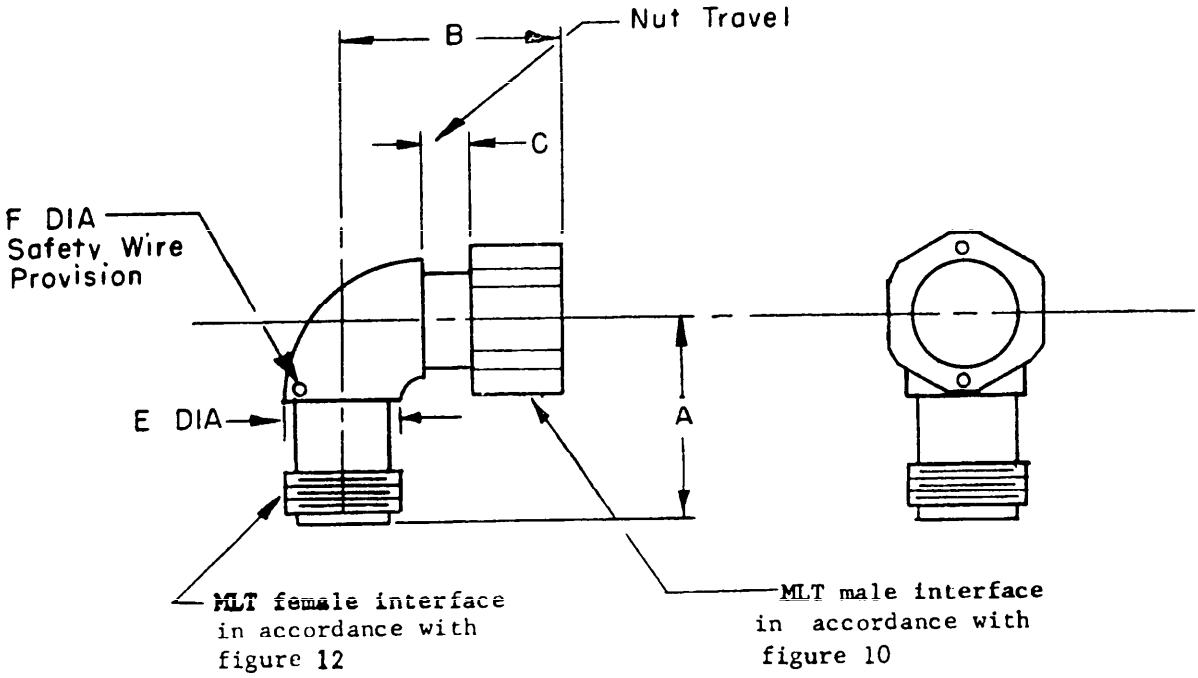
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	as required	note 1	as required	note 1
B	3.0	±0.25	76.2	±6.35
C	3.75	±0.06	95.25	±1.6
D	4.0	±0.06	101.6	±1.6
E	1.5	max	38.1	max
F	2.312	±0.06	58.72	±1.6
G	0.5	±0.03	12.7	±0.8

NOTES:

1. Tolerance for over-all cable length is ±0.25 inch up to 100 inches, and an additional ±0.25 inch for each additional 100 inches in excess of the first 100 inches.
2. Tags A and B to identify mating connector.

FIGURE 4. MLT cable assembly (Type D).

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.250	±.125	57.15	±3.200
B	2.437	±.063	61.91	±1.600
C	0.500	±.031	12.7	±0.8
D	0.750	min	19.05	min
E	1.500	max	38.1	max
F	0.064	+.004 -.001	1.626	+1.0 -.025

FIGURE 5. Right angle adapter.

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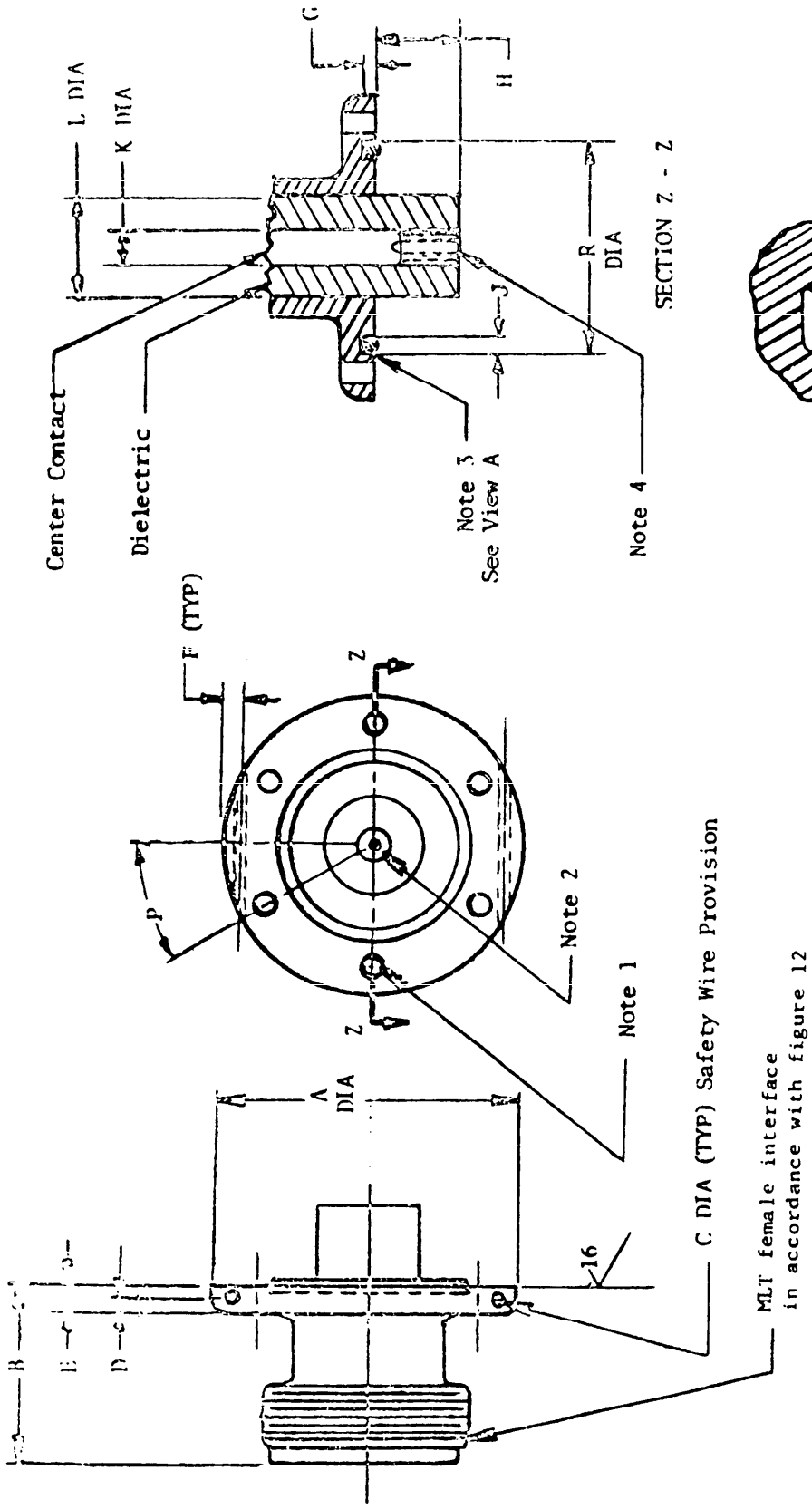


FIGURE 6 (Sheet 1 of 2). Panel receptacle.

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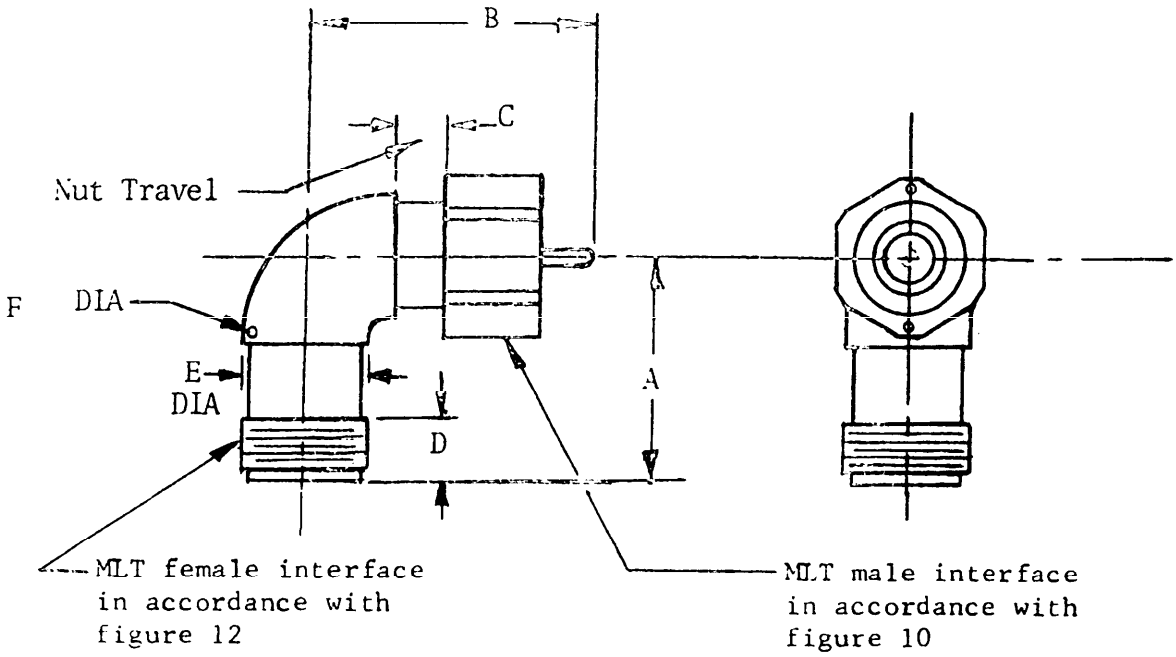
CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	1.88	±0.016	47.625	±0.4
B	1.156	±0.031	29.369	±0.8
C	0.064	+0.004 -0.001	1.626	+0.10 -0.025
D	0.093	±0.031	2.381	±0.8
E	0.187	±0.005	4.750	±0.127
F	0.125	±0.016	3.175	±0.4
G	0.047	+0.004 - 000	1.194	+0.10 - 000
H	0.5	±0.031	12.70	±0.8
J	0.082	+0.006 - 000	2.083	+0.152 - 000
K	0.220	±0.001	5.588	±0.025
L	0.624	±0.001	15.850	±0.025
M	0.010	max	.254	max
N	5°	max		
P	30°	±30'		
R	1.129	+0.000 -0.010	28.677	+ 000 -0.254

NOTES:

1. DIA. - 0.166 (4.2164) thru 6 holes equally spaced on 1.500 (38.1) ±0.005 (0.127) BCD within 0.008 (0.20) of true position.
2. 4-40UNC-2B, 0.375 (9.525), deep in center contact. Center within 0.002 (0.05) of receptacle center line.
3. 'O' ring - to be suited to application.
4. Center contact to be bonded in place, flush to 0.004 (0.10) under dielectric

FIGURE 6 (Sheet 2 of 2). Panel receptacle.

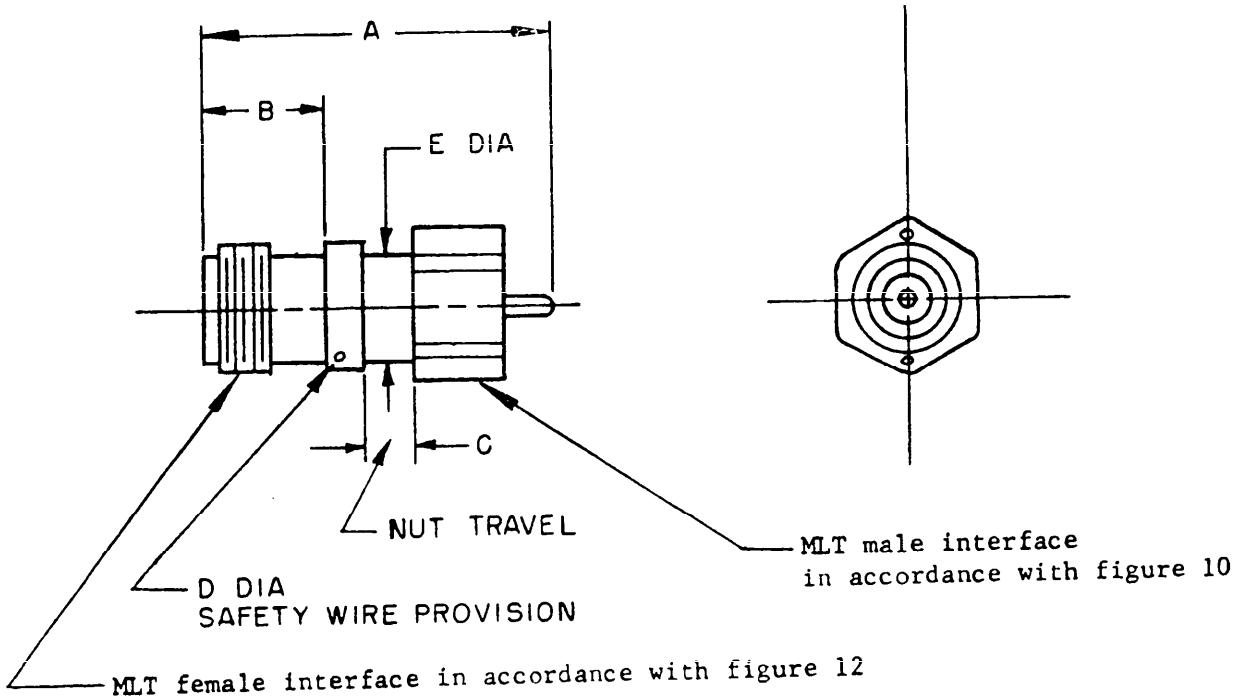
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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.25	±0.125	57.15	±3.2
B	0.875	±0.125	73.025	±3.2
C	0.5	±0.031	12.7	±0.8
D	0.75	min	19.05	min
E	1.50	max	38.1	max
F	0.064	+ .004 - .001	1.6256	+ .10 - .025

FIGURE 7. Right angle transition adapter.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	3.625	±0.125	92.08	±3.2
B	0.75	min	19.05	min
C	0.5	±0.031	12.7	±0.8
D	0.064	+ .004 - .001	1.626	+ .10 - .025
E	1.5	max	38.1	max

FIGURE 8. Straight transition adapter.

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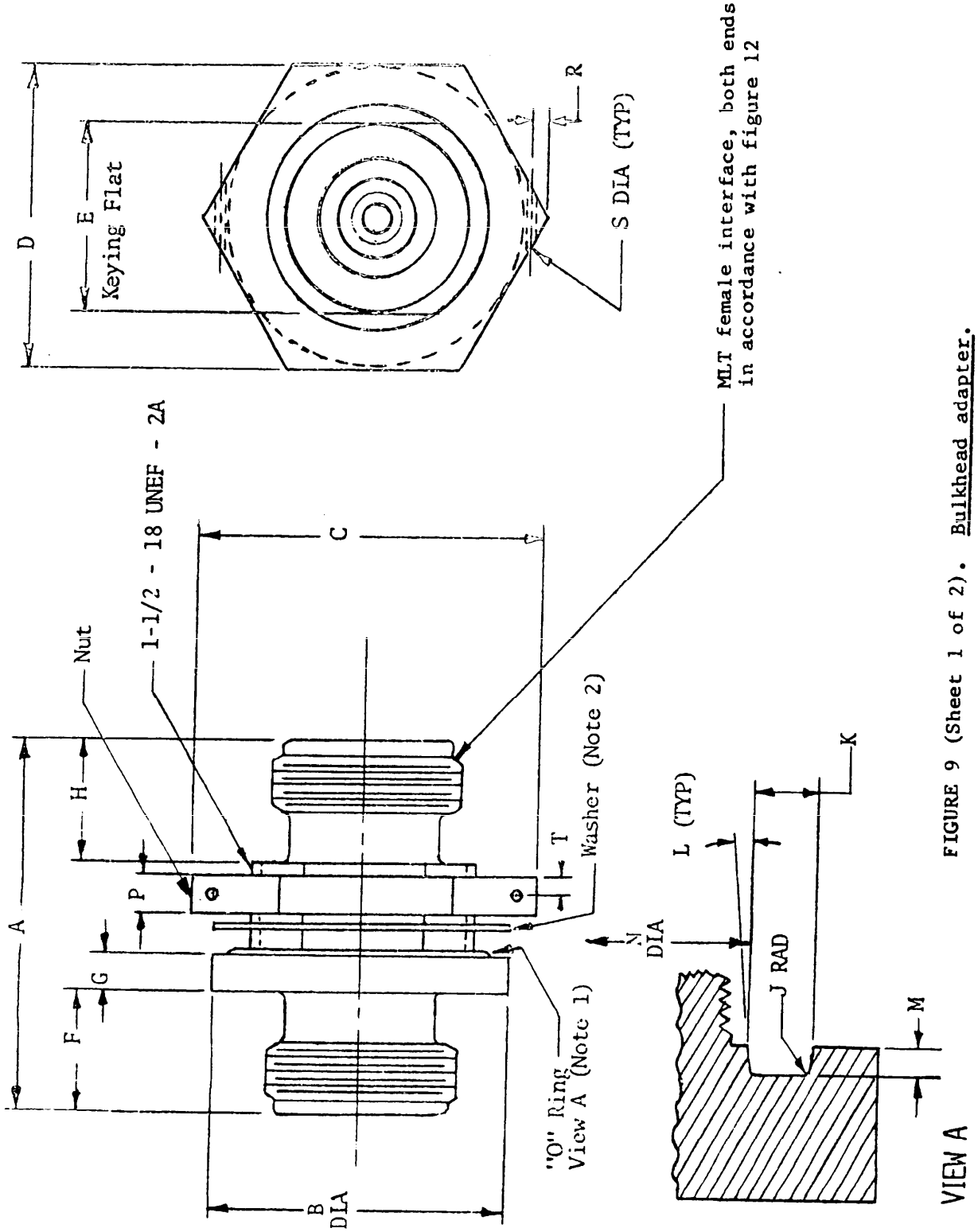


FIGURE 9 (Sheet 1 of 2). Bulkhead adapter.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	2.375	+0.031 - 0	60.325	+0.8 -0.0
B	2.0	+0.031 - 0	50.8	+0.8 -0.0
C	2.309	+0.000 - .01	58.649	+0.0 -0.25
D	2.0	±0.016	50.8	±0.4
E	1.250	±0.001	31.75	±0.025
F	0.75	+0.016 -0.00	19.05	+0.4 -0.0
G	0.25	±0.016	6.35	±0.4
H	0.75	+0.016 -0.00	19.05	+0.4 -0.0
J	0.010	max	0.25	max
K	0.112	±0.002	2.845	±0.05
L	5°	max		
M	0.048	±0.002	1.219	±0.05
N	1.562	+0.016 -0.00	39.688	+0.4 -0.0
P	0.25	±0.016	6.35	±0.4
R	0.125	±0.016	3.175	±0.4
S	0.064	+0.004 -0.001	1.626	+0.100 -0.025
T	0.125	±0.016	3.175	+0.4

NOTES:

1. 'O' ring - 1.77 (44.96) OD x 1.57 (39.88) ID x 0.100 DIA. cross section.
2. Washer - 2.0 (50.8) OD x 1.5312 (38.89) ID x 0.0625 (1.6) thick.

FIGURE 9 (Sheet 2 of 2). Bulkhead adapter.

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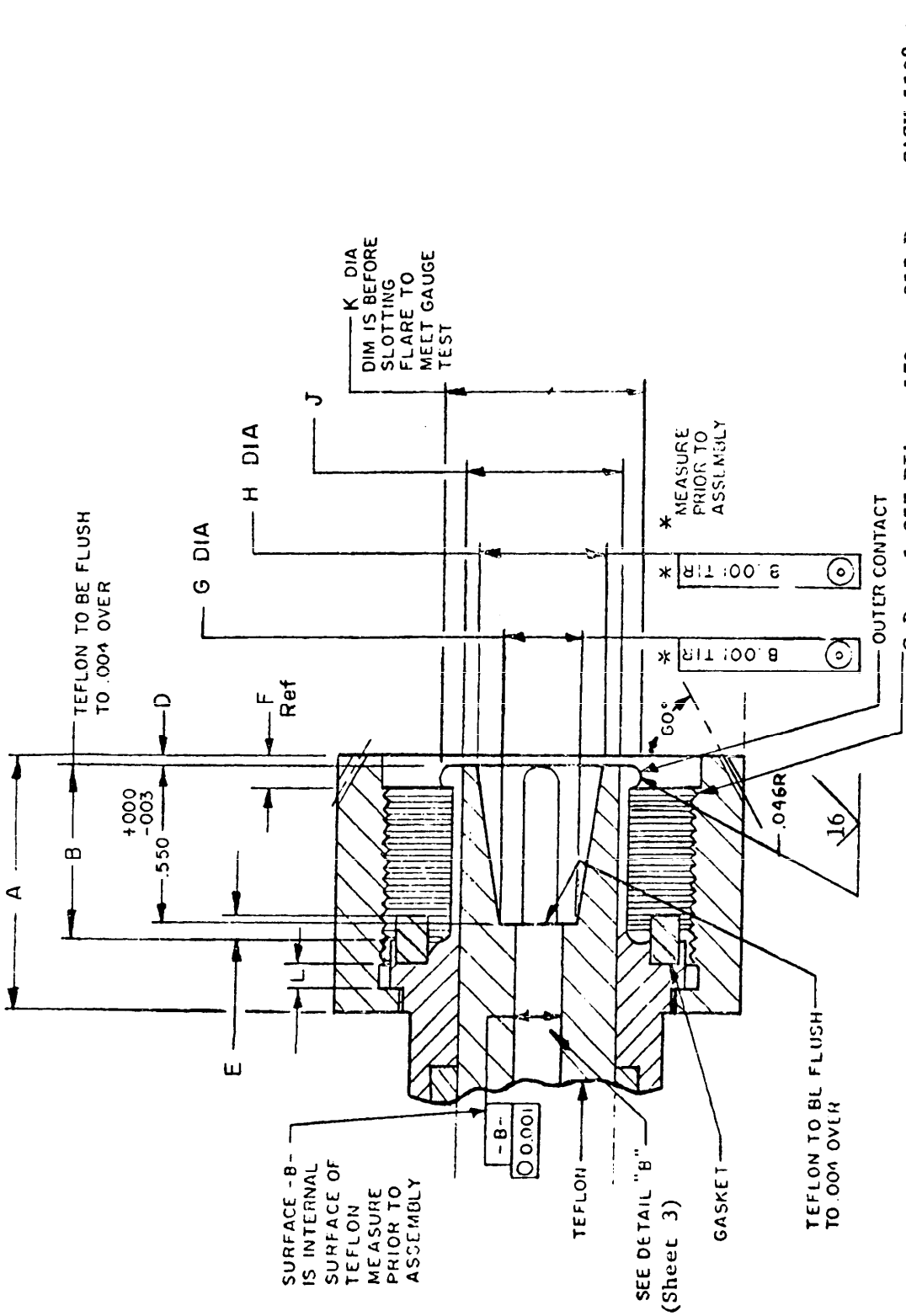


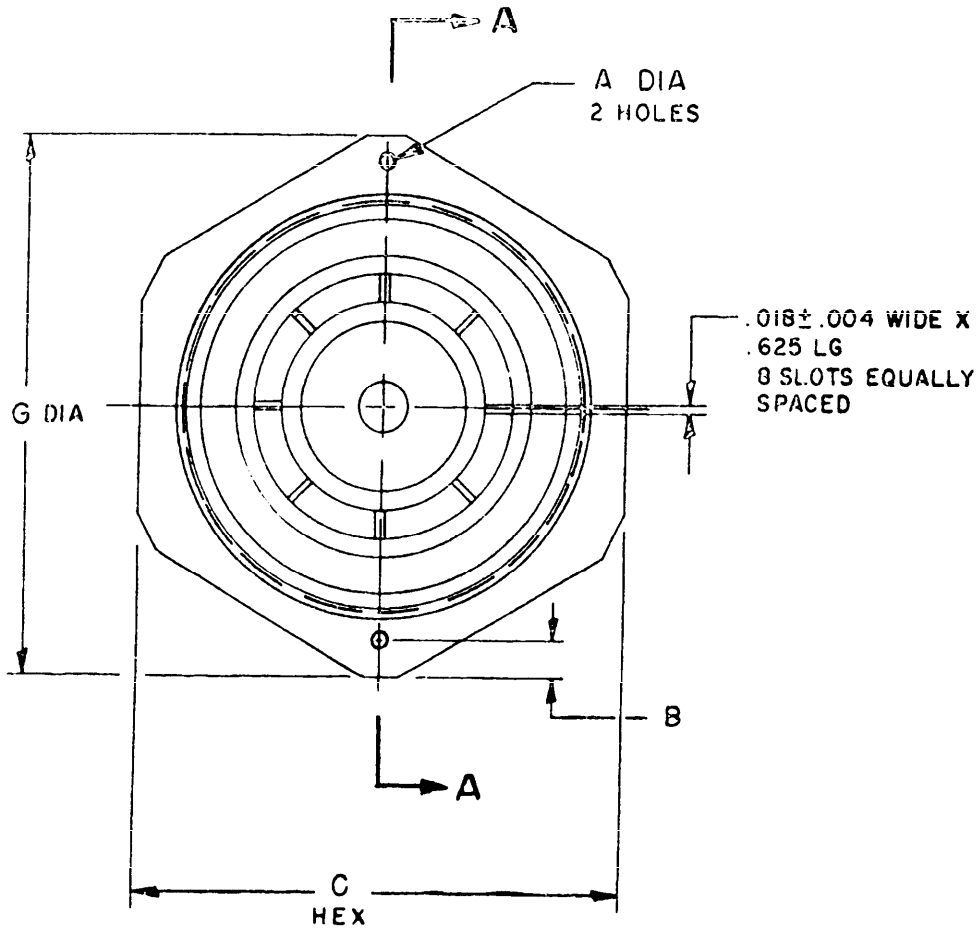
FIGURE 10 (Sheet 1 of 4). MLT male interface dimensions.

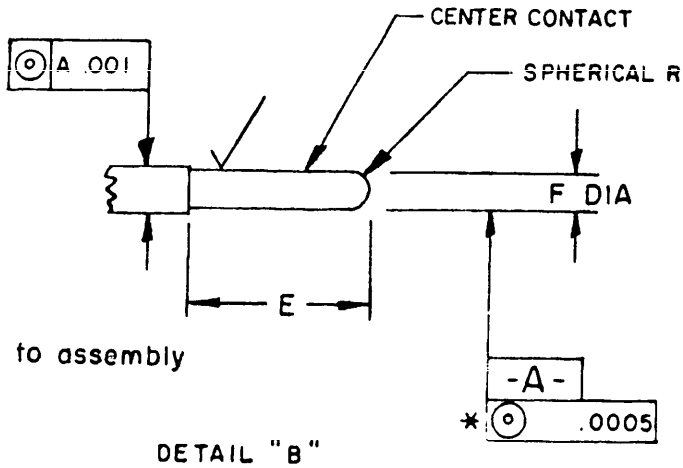
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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	1.0	±0.005	25.400	±0.127
B	0.669	±0.010	16.993	±0.254
C	0.550	+0.000 -0.003	13.970	+0.000 -0.003
D	0.031	±0.015	.787	±0.381
E	0.109	±0.005	2.769	±0.127
F	0.130	±0005	3.302	±0.127
G	0.300	+0.002 -0.000	7.620	+0.051 -0.000
H	0.500	+0.002 -0.000	12.700	+0.051 -0.000
J	0.625	+0.001 -0.002	15.875	+0.025 -0.051
K	0.790	±0.001	20.066	±0.025
L	0.9375	±0.016	23.813	±0.406

FIGURE 10 (Sheet 2 of 4). MLT male interface dimensions.

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May be measured prior to assembly

DETAIL "B"

FIGURE 10 (Sheet 3 of 4). MLT male interface dimensions (Detail "B").

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.064	+0.004 -0.001	1.626	+0.102 -0.025
B	0.109	±0.005	2.769	±0.125
C	1.50	±0.015	38.100	±0.381
D	1.625	±0.016	41.275	±0.337
E	0.550	+0.000 -0.005	13.970	+0.000 -0.127
F	0.150	±0.001	3.810	±0.025
G	1.625	±0.016	41.275	±0.337

FIGURE 10 (Sheet 4 of 4). MLT male interface dimensions (Detail "B").

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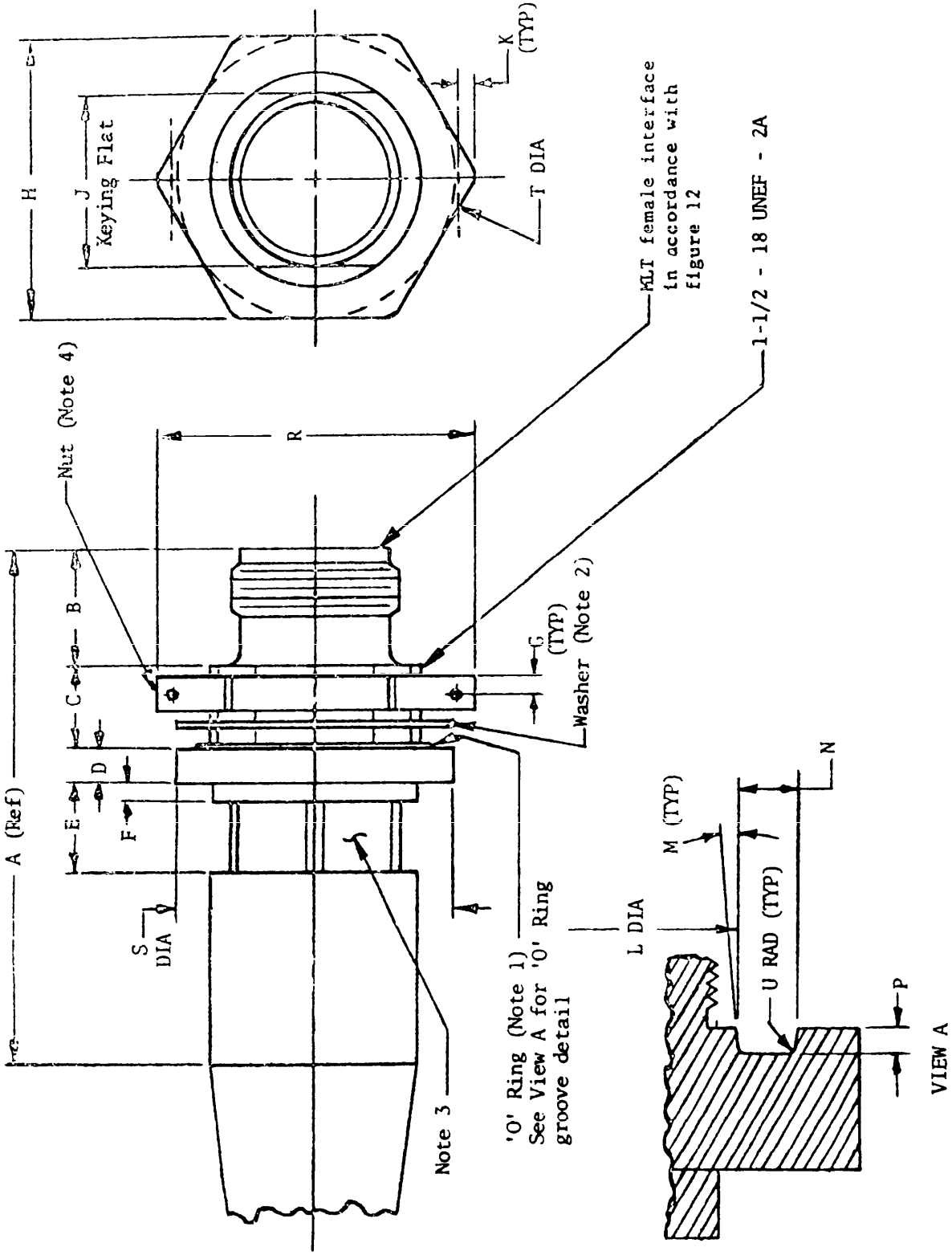


FIGURE 11 (Sheet 1 of 2). MLT female bulkhead connector.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	4.0	max	101.6	max
B	0.8125	+0.016 -0	20.63	+0.4 -0.0
C	0.63	±0.031	15.875	±0.8
D	0.25	±0.031	6.35	±0.8
E	0.63	±0.031	15.875	±0.8
F	0.13	±0.031	3.175	±0.8
G	0.13	±0.016	3.175	±0.4
H	1.50	±0.016	38.1	±0.4
J	1.250	±0.001	31.75	±0.025
K	0.1	±0.015	2.54	± .381
L	1.38	+0.016 -0	39.688	+0.4 -0.0
M	5°	max		
N	0.112	±0.002	2.845	±0.05
P	0.048	±0.002	1.219	±0.05
R	2.309	+0.0 -0.01	58.649	+0.0 -0.25
S	2.0	+0.031 -0.0	50.8	+0.8 -0.0
T	0.064	+0.004 -0.001	1.626	+0.100 -0.025
U	0.010	max	0.25	max

NOTES:

1. 'O' ring - 1.77 (44.96) OD x 1.57 (39.88) ID x 0.100 (2.5) DIA. cross section.
2. Washer - 2.0 (50.8) OD x 1.5312 (38.89) ID x 0.0625 (1.6) thick.
3. Hex flat - 1.250 (31.75).
4. Nut - 0.25 (6.35) thick.

FIGURE 11 (Sheet 2 of 2). MLT female bulkhead connector.

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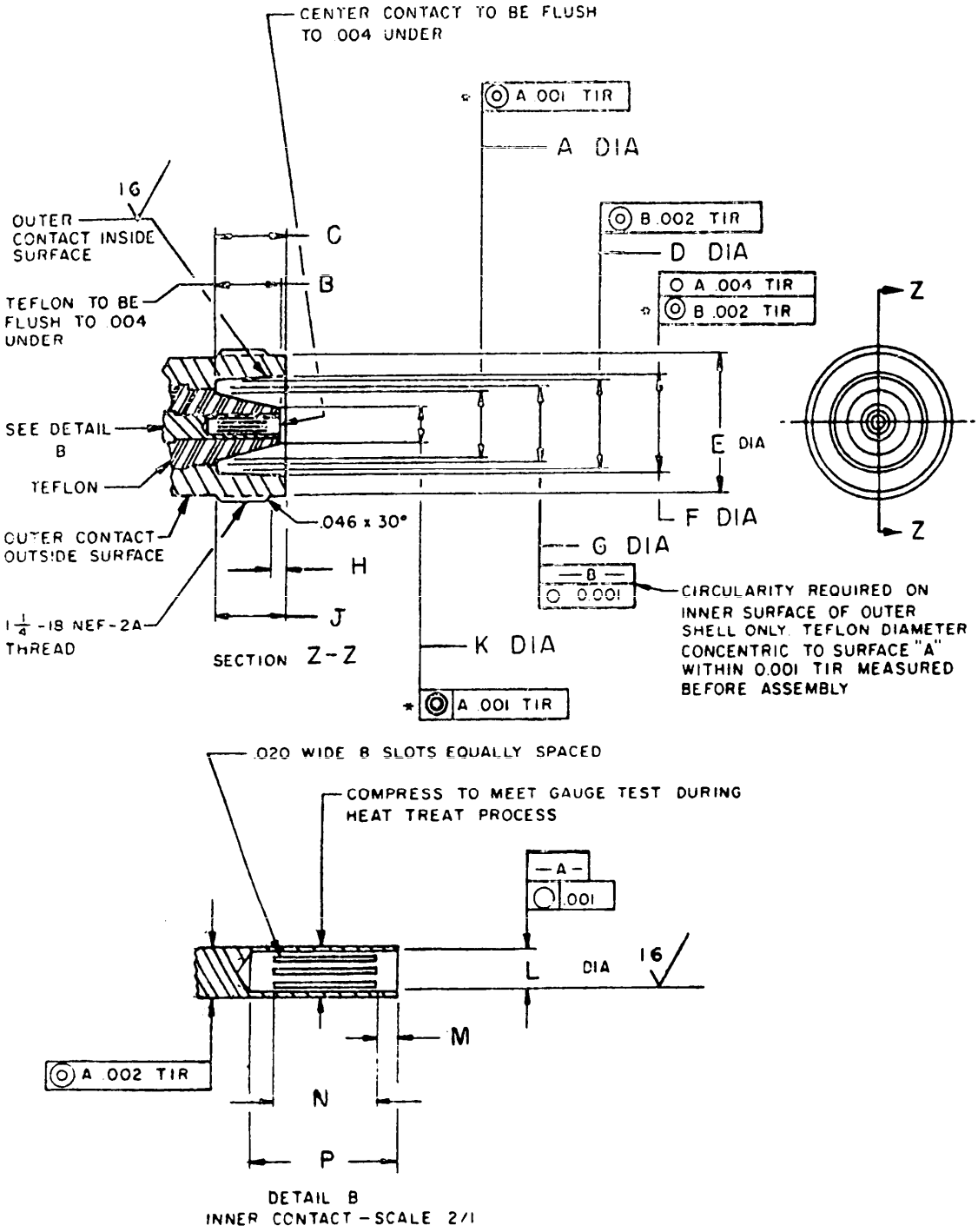


FIGURE 12 (Sheet 1 of 2). MLT female interface dimensions.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.504	+0.000 -0.002	12.802	+0.000 -0.051
B	0.550	+0.003 -0.000	13.970	+0.076 -0.000
C	0.580	±0.002	14.732	±0.051
D	0.790	±0.002	20.066	±0.051
E	1.150	±0.005	29.210	±0.127
F	0.802	±0.004	20.371	±0.102
G	0.624	±0.001	15.850	±0.025
H	0.140	±0.005	3.556	±0.127
J	0.562	±0.005	14.275	±0.127
K	0.304	+0.000 -0.002	7.722	+0.000 -0.051
L	0.154	±0.001	3.912	±0.025
M	0.080	±0.005	2.032	±0.127
N	0.437	±0.005	11.100	±0.127
P	0.625	±0.005	15.875	±0.127

FIGURE 12 (Sheet 2 of 2). MLT female interface dimensions.

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3.5.1.2 VSWR. The VSWR at the input to the coaxial assemblies shall not exceed the values specified in table II herein.

3.5.1.3 Insertion loss

3.5.1.3.1 Cable assemblies. The insertion loss of the cable assemblies shall not exceed the values specified in table III herein, as applicable.

3.5.1.3.2 Right angle adapter and right angle transition adapter. The insertion loss of the right angle adapter and the right angle transition adapter shall not exceed 0.12 dB at 5.5 GHz.

3.5.1.3.3 Panel receptacle, straight transition adapter, and bulkhead adapter. The insertion loss of the panel receptacle, the straight transition adapter, and the bulkhead adapter shall not exceed 0.1 dB at 5.5 GHz.

3.5.1.4 Corona

3.5.1.4.1 Cable assemblies, right angle adapter, panel receptacle, and bulkhead adapter. The corona extinction point of cable assemblies, the right angle adapter, the panel receptacle, and the bulkhead adapter shall not be less than 7000 volts rms.

3.5.1.4.2 Right angle transition adapter and straight transition adapter. The corona extinction point of the right angle transition adapter and the straight transition adapter shall not be less than 5000 volts rms.

3.5.1.5 High potential. The high potential test voltage shall be 10,000 volts rms.

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TABLE II. VSWR values.

ASSEMBLY TYPE	VSWR	FREQUENCY (GHz)	CW POWER (Watts)
CG-3795/U (Type A)	1.20:1	0.03 to 5.5	1000
CG-3792/U (Type B)	1.25:1	0.03 to 5.5	1000
CG-3802/U (Type C)	1.20:1	0.03 to 5.5	1000
CG-3803/U (Type D)	1.25:1	0.03 to 5.5	1000
RIGHT ANGLE ADAPTER	1.15:1	0.03 to 5.5	1000
PANEL RECEPTACLE	1.15:1	0.03 to 5.5	1000
RIGHT ANGLE TRANSITION ADAPTER	1.15:1	0.03 to 5.5	800
STRAIGHT TRANSITION ADAPTER	1.15:1	0.03 to 5.5	800
BULKHEAD ADAPTER	1.15:1	0.03 to 5.5	1000

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TABLE III. Insertion loss for cable assemblies.

FREQUENCY (GHz)	ATTENUATION (W)	LOSSES IN dB OF CONNECTORS (Y)	(Z)
0.10	1.00	0.027	0.047
0.30	1.85	0.030	0.050
1.00	4.10	0.035	0.055
2.00	6.60	0.040	0.060
3.00	8.80	0.045	0.065
4.00	10.90	0.050	0.070
5.00	12.60	0.053	0.083
5.50	13.50	0.055	0.080

COMPUTATION: The insertion loss for cable assemblies shall be computed as follows for each frequency shown above:

For Types A and C: $$\text{Insertion loss in dB} = \frac{W \times L}{100} + 2Y$$

For Types B and D: $$\text{Insertion loss in dB} = \frac{W \times L}{100} + (Y + Z)$$

WHERE:

W = Attenuation in dB per 100 feet
L = Lehgth in feet
Y = Insertion loss of straight connector
Z = Insertion loss of right angle connector.

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3.5.1.6 Altitude compliance

3.5.1.6.1 Corona requirements. At the altitude specified in MIL-C-87104, the corona extinction point of the coaxial assemblies shall not be less than the following:

a. Cable assemblies, right angle adapter, panel receptacle, and bulkhead adapter: 5000 volts rms

b. Right angle transition adapter and straight transition adapter: 4000 volts rms.

3.5.1.6.2 High potential requirements. At the altitude specified in MIL-C-87104, the high potential test voltage for coaxial assemblies shall be as follows:

a. Cable assemblies, right angle adapter, panel receptacle, and bulkhead adapter: 8000 volts rms

b. Right angle transition adapter and straight transition adapter: 7500 volts rms.

3.5.1.7 PHC. The coaxial assemblies shall withstand the CW power specified in table II herein, as applicable, and 20,000 watts of peak power, at a duty cycle of 5 percent, at 5.5 GHz.

3.5.1.8 R-f leakage. The r-f suppression capability shall be 90 dB/ft.

3.5.1.9 VP

3.5.1.9.1 Cable assemblies. The velocity of propagation for cable assemblies shall be as specified in table IV herein.

3.5.1.9.2 Right angle adapter, panel receptacle, right angle transition adapter, straight transition adapter, and bulkhead adapter. The velocity of propagation for the right angle adapter, the panel receptacle, the right angle transition adapter, the straight transition adapter, and the bulkhead adapter shall not be less than 69 percent.

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TABLE IV. Velocity of propagation for cable assemblies.

COMPUTATION: The velocity of propagation (VP) for cable assemblies shall not be less than the values as calculated in the equation below:

For Types A and C: $$VP = \frac{(77.0 \times L) - 37.8 \text{ (in. \%)}}{L}$$

For Types B and D: $$VP = \frac{(77.0 \times L) - 52.5 \text{ (in. \%)}}{L}$$

WHERE:

L = Inches
Quotient = Percent of free space propagation.

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3.5.2 Mechanical requirements. Mechanical values for the coaxial assemblies shall be as specified herein.

3.5.2.1 Tensile load stress force. The coaxial assemblies shall withstand a tensile load stress force of 150 pounds.

3.5.2.2 Torque values

3.5.2.2.1 Cable assemblies. Cable assemblies shall withstand a torque force of 50 inch-pounds.

3.5.2.2.2 Rigid assemblies and connectors. Rigid assemblies and connectors shall withstand a torque force of 300 inch-pounds.

3.5.2.2.3 Coupling nut. The coupling nut shall be torqued to 100 inch-pounds prior to performing the torque test.

3.5.2.3 Minimum bend radius. Unless otherwise indicated on the figures herein, the minimum bend radius for cable assemblies shall be 3 inches.

3.5.2.4 Outer diameter of cable. The outer diameter of coaxial cable shall be a nominal 0.75 inch.

3.5.3 Environmental conditions

3.5.3.1 Flow test weight. The flow test weight shall be 15 pounds.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions of MIL-C-87104 shall apply to this specification, with additions and exceptions specified herein. When the general specification and this specification conflict, this specification shall govern.

4.2 Quality conformance test program. A coaxial assembly that is procured with lesser frequency coverage than that specified herein shall be subjected to quality conformance testing in the frequency range of 0.03 GHz to the maximum frequency required for that assembly.

4.3 Methods of inspection. The required specific testing values for electrical properties, mechanical requirements, and environmental conditions shall be as specified herein under 3.5.1, 3.5.2, and 3.5.3, respectively. Additional testing requirements shall be as specified herein.

4.3.1 Insertion loss frequencies. The insertion loss tests shall be conducted at each of the frequencies specified under 3.5.1.3 herein.

4.3.2 Interface gauging tests

4.3.2.1 Procedure I. Procedure I shall be an acceptance test for female MLT type connectors.

4.3.2.1.1 Test apparatus. Gauge pins shall be in accordance with figures 13, 14, and 15 herein. Associated indicator light circuits shall also be required.

4.3.2.1.2 Procedural steps. Procedure I shall be conducted in accordance with the following steps:

a. Step 1: Connect pin A to an indicator light circuit.

b. Step 2: Insert pin A into the center contact until it bottoms.

c. Step 3: Slowly rotate pin A through 360° while observing the indicator light.

d. Step 4: Remove pin A and connect the indicator light circuit to pin B.

e. Step 5: Insert pin B into the center contact until it bottoms.

f. Step 6: Slowly rotate pin B through 360° with translation while observing the indicator light.

g. Step 7: Remove pin B.

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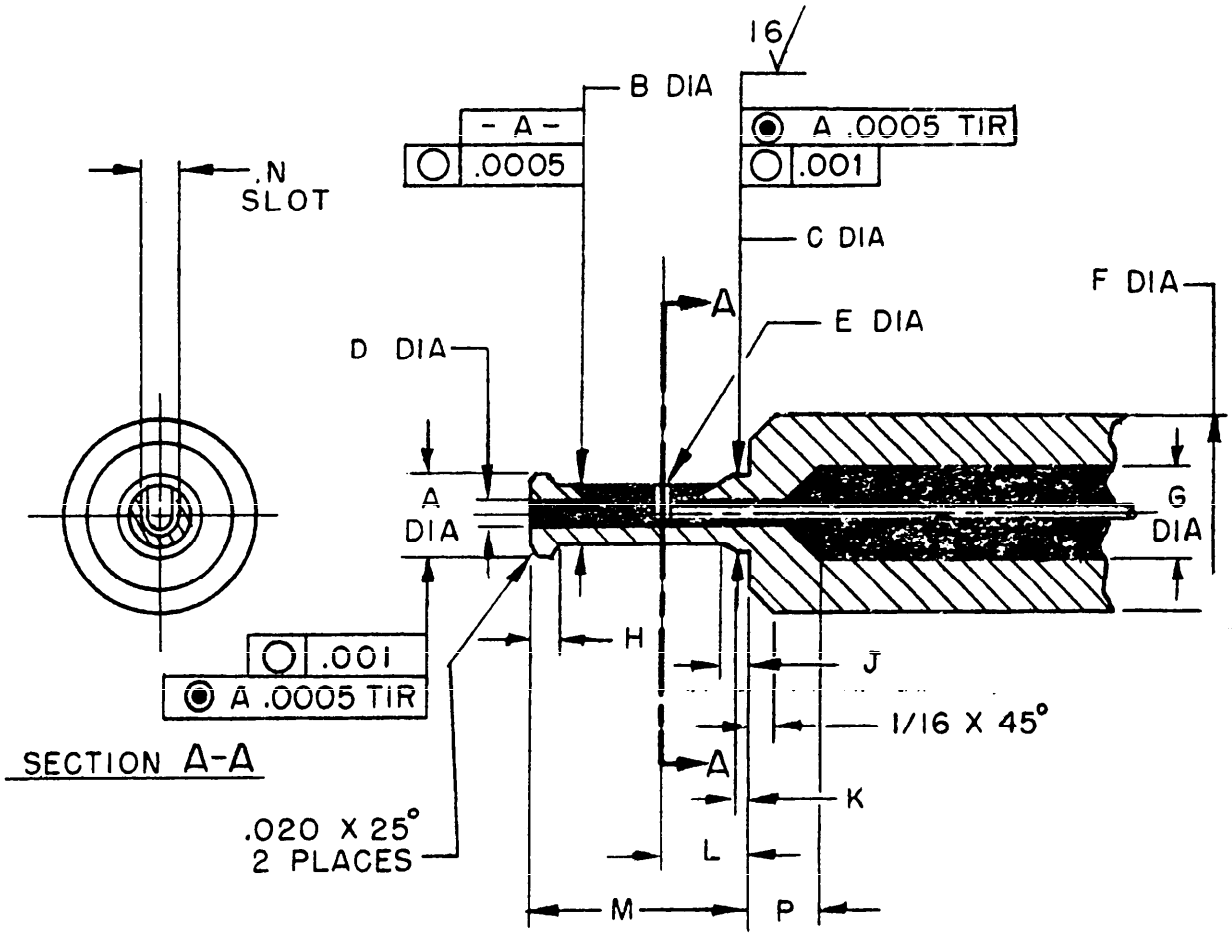


FIGURE 13 (Sheet 1 of 2). MLT pin gauge A.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.189	±0.0005	4.801	±0.013
B	0.184	+0.0000 -0.0005	4.674	+0.0000 -0.013
C	0.189	±0.0005	4.801	±0.127
D	0.062	+0.004 -0.001	1.575	+0.102 -0.025
E	0.062	±0.002	1.575	±0.051
F	0.5	±0.016	12.700	±0.406
G	0.25	±0.016	6.350	±0.406
H	0.063	±0.005	1.600	±0.127
J	0.094	±0.016	2.388	±0.406
K	0.75	±0.016	19.050	±0.406
L	0.25	±0.016	6.350	±0.406
M	0.594	±0.005	15.088	±0.127
N	0.075	±0.0005	1.905	±0.013
P	0.188	±0.016	4.775	±0.406

FIGURE 13 (Sheet 2 of 2). MLT pin gauge A.

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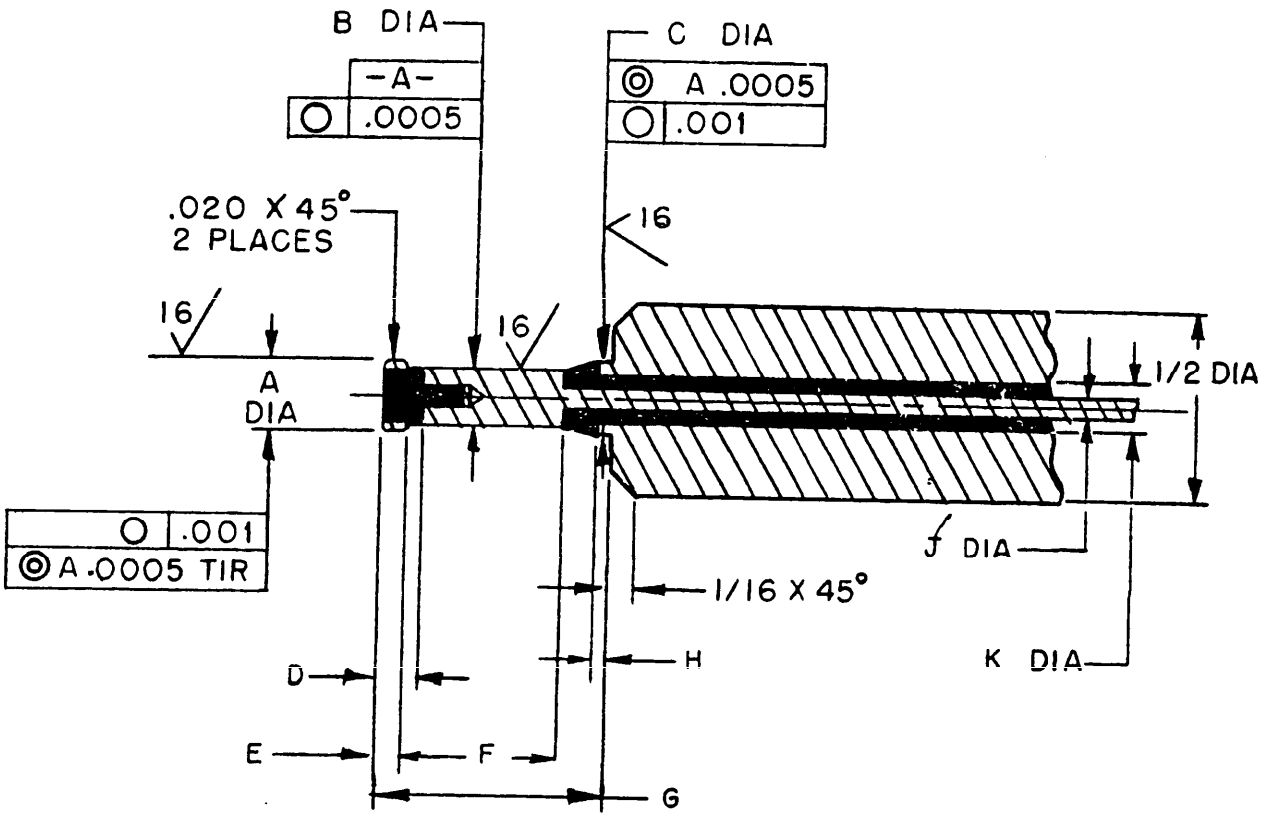


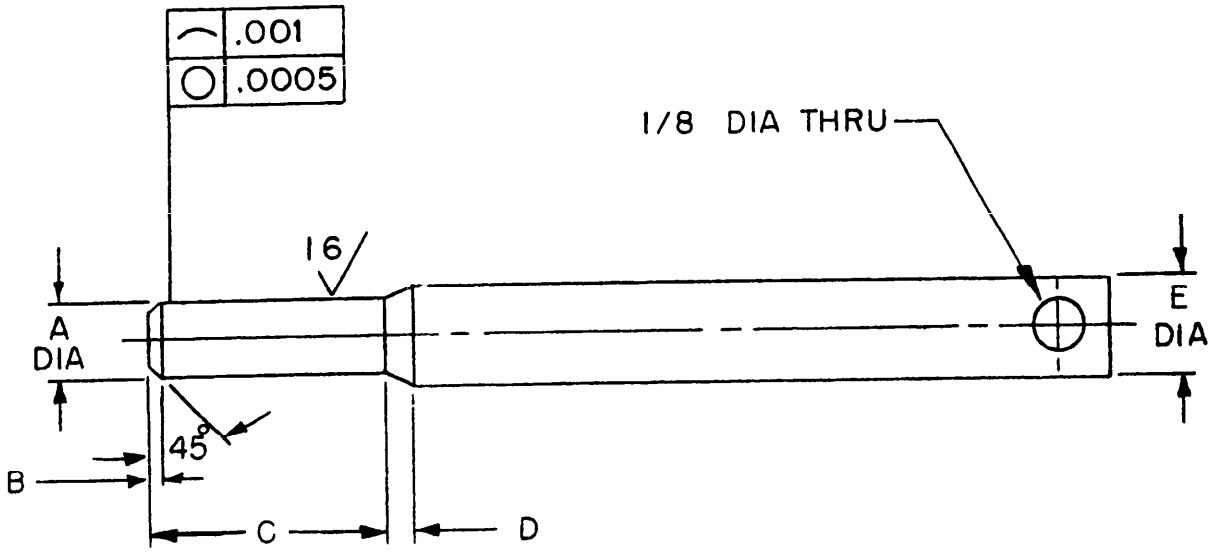
FIGURE 14 (Sheet 1 of 2). MLT pin gauge B.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:53Z
Check the source to verify that this is the current version before use.

CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.189	±0.0005	4.801	±0.013
B	0.178	+0.0000 -0.0005	4.521	+0.000 -0.013
C	0.189	±0.0005	4.801	±0.013
D	0.125	±0.005	3.175	±0.127
E	0.063	±0.005	1.600	±0.127
F	0.437	±0.005	11.100	±0.127
G	0.594	±0.005	15.088	±0.127
H	0.047	±0.016	1.194	±0.406
J	0.063	±0.005	1.600	±0.127
K	0.125	±0.005	3.175	±0.127

FIGURE 14 (Sheet 2 of 2). MLT pin gauge B.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.187	±0.0005	4.750	±0.0127
B	0.031	±0.016	0.787	±0.406
C	0.594	±0.005	15.088	±0.127
D	0.063	±0.016	1.600	±0.406
E	0.250	±0.01	6.350	±0.25
F	45°	±30'		

FIGURE 15. MLT pin gauge C.

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h. Step 8: Insert pin C into the center contact.

i. Step 9: Attach a spring scale, or other appropriate means of measuring tension, to pin C.

j. Step 10: Slowly withdraw pin C from the center contact. Record the number of pounds of force required to withdraw pin C.

k. Step 11: Repeat step 10, two additional times.

4.3.2.1.3 Accept-reject criteria (procedure I). The female MLT type connector shall be considered to have passed the interface gauging test if:

a. During step 3, the light remains lit at all angles of rotation, or if the light has an unlighted condition for less than 15° of rotation.

b. During step 6, there is a position of no contact between pin B and the center contact as evidenced by the unlighted condition of the indicator lamp.

c. During steps 10 and 11, the force required to withdraw pin C is 6 ±2 pounds.

4.3.2.2 Procedure II. Procedure II shall be an acceptance test for male MLT type connectors.

4.3.2.2.1 Test apparatus. Ring and fit gauges shall be in accordance with figures 16, 17, and 18 herein. Associated indicator light circuits shall also be required.

4.3.2.2.2 Procedural steps. Procedure II shall be conducted in accordance with the following steps:

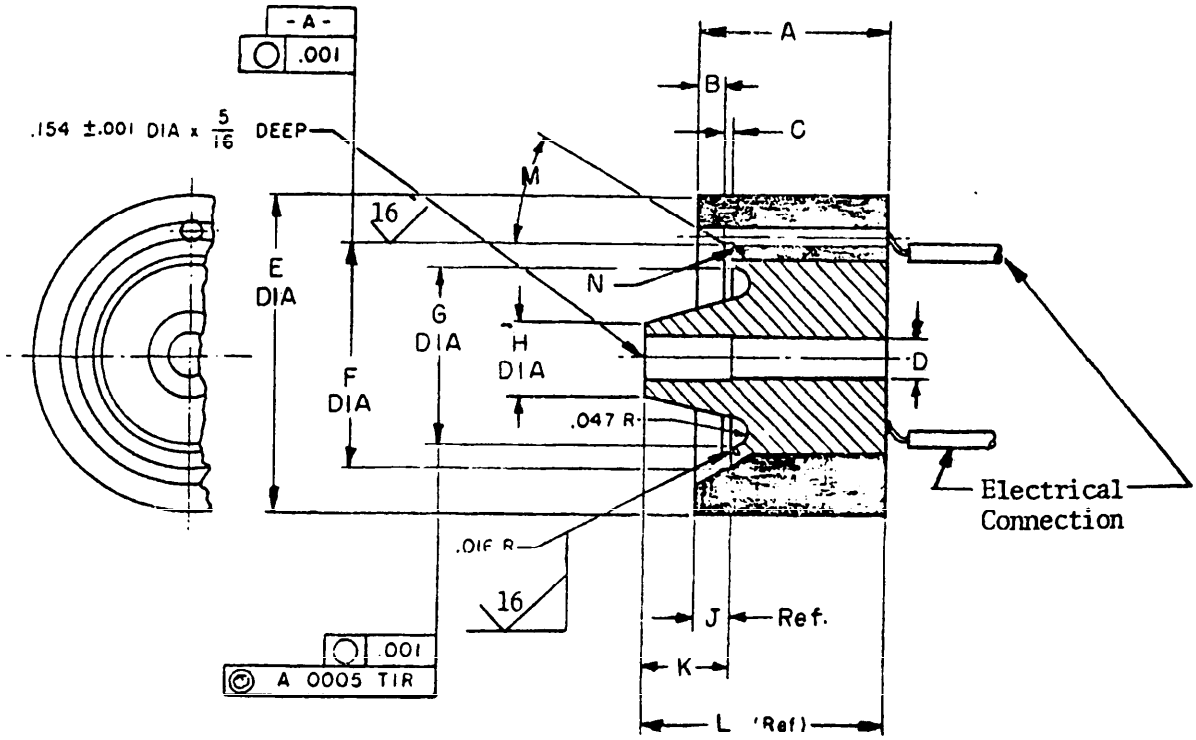
a. Step 1: Connect ring A to an indicator light circuit.

b. Step 2: Place ring A over the outer contact until it seats.

c. Step 3: Keeping ring A firmly seated, slowly rotate the ring gauge through 360° while observing the indicator light.

d. Step 4: Remove ring A and connect the indicator light circuit to ring B.

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NOTES:

1. Shaded areas denote nonconducting material.
2. Surface roughness 125 on all machined surfaces not indicated.

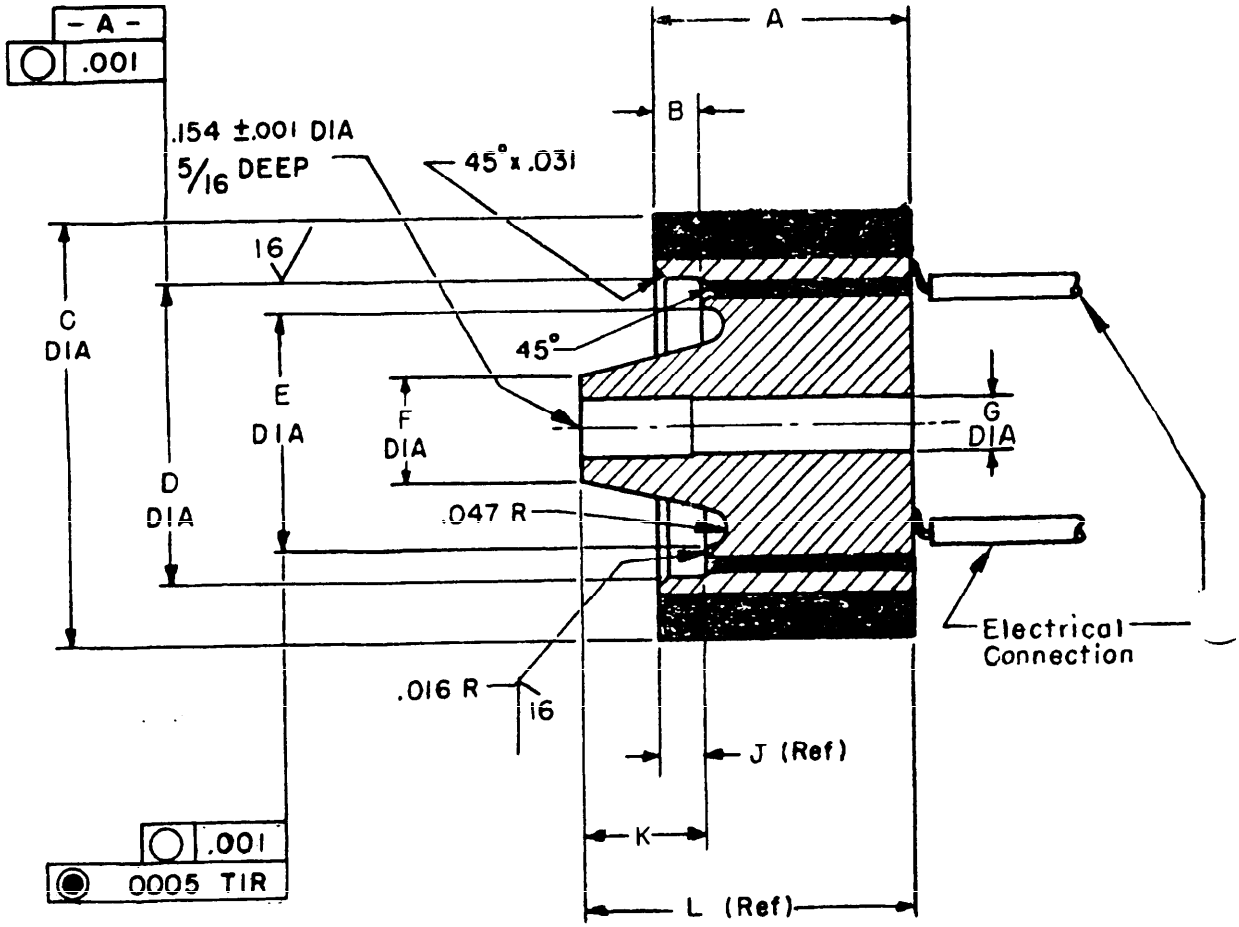
FIGURE 16 (Sheet 1 of 2). MLT ring gauge A.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.688	±0.016	17.463	±1.590
B	0.125	±0.016	3.175	±1.590
C	0.055	max	1.397	max
D	0.140	+0.005 -0.001	3.556	+0.127 -0.025
E	1.125	±0.016	28.575	±1.590
F	0.794	+0.000 -0.001	20.168	+0.000 -0.025
G	0.622	+0.000 -0.001	15.799	+0.000 -0.025
H	0.250	±0.005	6.350	±0.127
J	0.156	±0.063	3.969	±1.588
K	0.312	±0.005	7.925	±0.127
L	0.844	±0.063	21.431	±0.127
M	30°	±30'		
N	45°	±30'		

FIGURE 16 (Sheet 2 of 2). MLT ring gauge A.

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NOTES:

1. Shaded areas denote nonconducting material.
2. Surface roughness 125 on all machined surfaces not indicated.

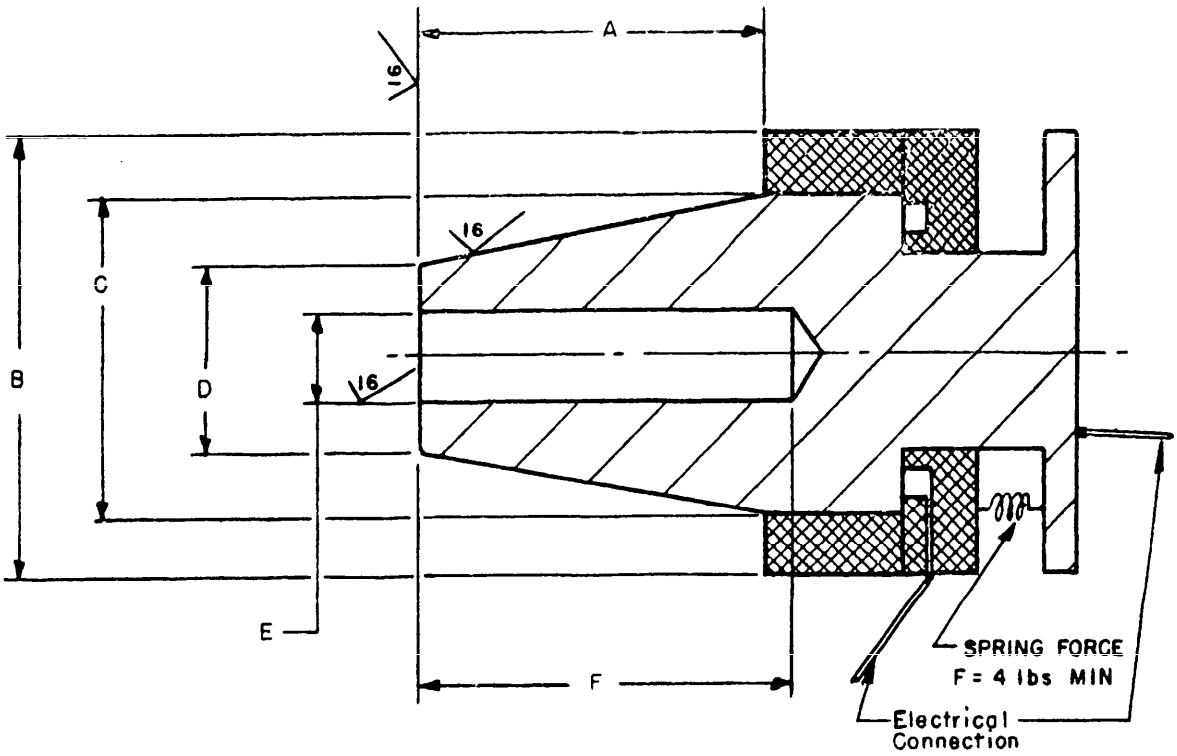
FIGURE 17 (Sheet 1 of 2). MLT ring gauge B.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.688	±0.016	17.475	±0.407
B	0.156	±0.016	3.962	±0.407
C	1.125	±0.016	28.575	±0.407
D	0.805	+0.000 -0.001	20.447	+0.000 -0.025
E	0.622	+0.000 -0.001	15.798	+0.000 -0.025
F	0.250	±0.005	6.350	±0.127
G	0.140	+0.005 -0.001	3.556	+0.127 -0.025
H	45°	±30'		
J	0.156	±0.016	3.962	±0.407
K	0.312	±0.005	7.925	±0.127
L	0.844	±0.016	21.438	±0.407

FIGURE 17 (Sheet 2 of 2). MLT ring gauge B.

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NOTES:

1. Shaded areas denote nonconducting material.
2. Surface roughness 125 on all machined surfaces not indicated.

FIGURE 18 (Sheet 1 of 2). MLT ring gauge C.

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CODE LETTER	INCHES		MM	
	DIMENSION	TOLERANCE	DIMENSION	TOLERANCE
A	0.5472	±0.0002	13.8989	±0.0051
B	0.620	±0.002	15.748	±0.051
C	0.4998	±0.0002	12.6949	±0.0051
D	0.2998	±0.0002	7.6149	±0.0051
L	0.1528	±0.0002	3.8811	±0.0051
F	0.60	±0.01	15.24	±0.26

FIGURE 18 (Sheet 2 of 2). MLT ring gauge C.

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e. Step 5: Place ring B over the outer contact until it seats.

f. Step 6: Keeping ring B firmly seated, slowly rotate the ring gauge with translation while observing the indicator light.

g. Step 7: Remove ring B and connect the indicator light circuit to ring C.

h. Step 8: Insert ring C into the connector until it seats.

i. Step 9: Press ring C over the center pin exerting a 12-pound force along the axis of pin C. Observe the indicator light.

4.3.2.2.3 Accept-reject criteria (procedure II). The male connector shall be considered to have passed the interface gauging test if:

a. During step 3, the indicator light remains lit through all degrees of rotation.

b. During step 6, there is a point of no contact between ring B and the outer contact as evidenced by the unlighted condition of the indicator lamp.

c. The indicator light is not lit during the application of the 12-pound force.

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5. PACKAGING

5.1 Section 5 of MIL-C-87104 shall apply.

6. NOTES

6.1 Section 6 of MIL-C-87104 shall apply.

6.1.1 The contract should specify the required information for tags A and B.

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